



RS004 Week 8 — English Worksheet

Topic: Side Collision + Varied Platforms (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

Until now the player only landed from above. This week walls **block from the side**, and bumping your **head** on a platform stops you and drops you back down. The trick: handle the **horizontal** move and the **vertical** move *separately*.

Keep these words handy: **side collision**, **horizontal / vertical**, **player.right** / **player.left**, **head bump**, **wall**.

1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
player.x = player.x + vx
for plat in platforms:
    if player.colliderect(plat):
        if vx > 0:
            player.right = plat.left
```

The player moves right into a wall. What does **player.right = plat.left** do?


```
elif vy < 0:
    player.top = plat.bottom
    vy = 0
```

vy < 0 means moving up. The player's head hits a platform. What happens next?


```
# 1. move horizontally, check horizontal collisions  
# 2. move vertically, check vertical collisions
```

Why do we handle horizontal and vertical movement in two separate steps?

2 · Edge Words

Match the line to what it lines up. A player runs into a platform.

1. `player.right = plat.left`
2. `player.left = plat.right`
3. `player.bottom = plat.top`
4. `player.top = plat.bottom`

- A. player landing on top of the platform
- B. player's head bumping the platform's underside
- C. player stopped at the platform's left wall (moving right)
- D. player stopped at the platform's right wall (moving left)

1 → __ 2 → __ 3 → __ 4 → __

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — Moving right into a wall should stop the player at the wall. Right now the player slides straight through.

```
player.x = player.x + vx
for plat in platforms:
    if player.colliderect(plat):
        vx = 0
```

Hint: setting `vx = 0` after moving doesn't undo the overlap. Snap the player's edge to the wall.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Head bumps and landings should both work. Right now a head bump is treated like a landing, so the player sticks to the ceiling.

```
player.y = player.y + vy
for plat in platforms:
    if player.colliderect(plat):
        player.bottom = plat.top
        vy = 0
```

Hint: check the *direction* of `vy`. Down (`> 0`) is landing, up (`< 0`) is a head bump.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Horizontal and vertical checks should be separate. Right now they happen in one move, and the player jitters and gets stuck in corners.

```
player.x += vx
player.y += vy
for plat in platforms:
    if player.colliderect(plat):
        player.right = plat.left
        player.bottom = plat.top
```

Hint: move on X and resolve X first, *then* move on Y and resolve Y.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Design with Walls

Start from your working full collision.

1. Add one tall thin platform to act as a **wall**. Walk into it from both sides and confirm it blocks you.
Write its Rect .

2. Add a low platform you can bump your **head** on. Where did you place it?

3. **Stretch:** build a narrow corridor the player must walk through. Sketch the idea in words.

5 · Build & Show 📷

Design **your own** level using platforms of different sizes — include at least one wall the player bumps into from the side.

When it works, send a **photo or video** showing top, side, and head collisions, then explain what you did. Use these starters — write 4 to 6 sentences.

First, I split movement into horizontal and vertical because ...

When the player hits a wall on the right, I set ...

A head bump is when $vy < 0$, and I handle it by ...

My level has a wall / corridor at ...

One tricky moment was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video showing all three collision types. Teach it to someone new. Try to use these words: **side collision, wall, head bump, horizontal, vertical**.

1. Walk into a wall from the side and show it blocks you.
2. Jump up under a platform and bump your head.
3. Land on a platform from above.
4. Read one collision line out loud and explain which edge it lines up.

Write what you will say in your video. Plan it here first.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.